Why the Blockchain is So Secure

A blockchain has several built-in security features that make it attractive for purposes like land records, cryptocurrency transactions, etc. The security of personal data is a human right. A blockchain could be one of the methods of ensuring this.

Simply defined, a blockchain is, "A decentralised database containing sequential, cryptographically linked blocks of digitally signed asset transactions, governed by a consensus model." Blockchain technology is a peer-to-peer networked database governed by a set of rules. A blockchain represents a shift away from traditional trust agents and a move towards transparency. As a technological building block, it permits applications from a broad band of industries to take advantage of sharing, tracking, and auditing digital assets.

Blockchain is a disruptive technology because of its ability to digitise, decentralise, secure and incentivise the validation of transactions. A wide swathe of industries is evaluating blockchain to determine what strategic differentiators could exist for their businesses if they leverage it.

This technology has the potential to improve security, processes and systems in the financial services domain, in government and every sphere where accurate, tamperproof record-keeping is essential.

The disrupted industries will include financial services, healthcare, aviation, global logistics and shipping, transportation, music, manufacturing, security, media, identity, automotive, land use and government.

Industry adoption of blockchain platforms

Market research predicts that, by 2024, the global blockchain market is expected to be worth over US$ 20 billion. The use and adoption of blockchain technology is expanding at a rapid pace, all over the world.

The Republic of Georgia has declared that it will use blockchain technology to validate property-related government transactions. Countries like Sweden, Honduras and others are also developing such similar blockchain based systems, for enabling secured e-governance.

Gartner projects that the business value added by blockchain will grow to US$ 176 billion by 2025.

Recently, the Dubai government announced that it will put 100 per cent of its records pertaining to land registry on blockchain. Dubai Land Department (DLD), in fact, claims to be the first such government department anywhere in the world to adopt the blockchain for such high-level tasks.

The European Union's commercial research group, the European Innovation Council (EIC), has launched a programme to grant US$ 3.04 billion (2.7 billion Euros) to 1000 projects that are developing systems and solutions using blockchain technology.

Key characteristics of blockchain technology

Every transaction that records and stores is not labelled as a blockchain. The following are the main characteristics of blockchain.

- **Digital:** All the information on blockchain is digitised, thus eliminating the need for manual documentation.
- **Distributed:** Blockchain distributes control among all peers in the transaction chain, creating a shared infrastructure within an enterprise system. Participants independently validate information without a centralised authority. There is no